

Chapter 12, Question 6

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Question. Given the first M principal component score vectors $z_{i1}, \dots, z_{iM}$, explain, using equation (12.6), that each of the first M principal component loading vectors $\phi_{j1}, \dots, \phi_{jM}$ can be obtained by performing p separate least squares linear regressions, in which the scores are the predictors and one feature of the data matrix is the response.

Starting point

Equation (12.6) frames PCA as finding the scores (a_{im}) and loadings (b_{jm}) that together minimize the total squared reconstruction error:

$$\underset{A, B}{\text{minimize}} \sum_{j=1}^p \sum_{i=1}^n \left(x_{ij} - \sum_{m=1}^M a_{im} b_{jm} \right)^2$$

There are two unknowns here: the $n \times M$ score matrix A and the $p \times M$ loading matrix B . The question asks what happens once the scores are already known and fixed, leaving only the loadings still to find.

Splitting the sum by feature

With the scores z_{im} fixed, the only thing left to solve for is $b_{j1}, \dots, b_{jM}$. Here's the key observation: the outer sum runs over features $j = 1, \dots, p$, and each term in it only touches that one feature's own loadings — nothing from feature j ever mixes with feature k 's loadings. That means the whole problem falls apart into p separate, independent problems, one per feature:

$$\text{For each fixed } j : \underset{b_{j1}, \dots, b_{jM}}{\text{minimize}} \sum_{i=1}^n \left(x_{ij} - \sum_{m=1}^M z_{im} b_{jm} \right)^2$$

Why this is just least squares

Take a fixed feature j and look at what this problem actually looks like. x_{ij} , across all n observations, is nothing more than a single response variable — feature j 's column of values. And $\sum_m z_{im} b_{jm}$ is a linear combination of the M known scores $z_{i1}, \dots, z_{iM}$, with $b_{j1}, \dots, b_{jM}$ as the unknowns to solve for.

That's exactly a least squares regression: regress x_j (feature j 's values) on the M predictors $z_1, \dots, z_M$, no intercept needed since both the scores and the original data are centered at zero. Whatever coefficients come out of that regression — $\hat{b}_{j1}, \dots, \hat{b}_{jM}$ — are exactly $\phi_{j1}, \dots, \phi_{jM}$, the loadings for feature j .

Putting it together

The same argument works for every feature $j = 1, \dots, p$, and since none of the p sub-problems depend on each other, you get the full set of loadings by running p separate least squares regressions:

$$x_j \sim z_1 + z_2 + \dots + z_M \quad (\text{no intercept})$$

one regression per feature, reading the coefficients straight off as the loadings.

Why this actually makes sense

This is really just PCA viewed backwards from how it's usually taught. Normally you find the loadings first (the directions of max variance) and get the scores as a byproduct. But (12.6) doesn't actually care which one you solve for first — it's symmetric in A and B . So if you flip it around and fix the scores instead, asking "what loadings best reconstruct the data from these scores," you land on ordinary least squares, because that's all (12.6) really is: a sum of squared residuals. And minimizing a sum of squared residuals when the predictors are known and only the coefficients are unknown is, by definition, what least squares does.